Tut 2 $\quad 1-4x^2 = 1-(2x)^2$

Ex. 7.6

Q 48 $\quad \displaystyle\int \frac{dx}{\sqrt{1-4x^2}} = \int \frac{du}{2\sqrt{1-u^2}}$

$u = 2x$

$du = 2\,dx$

$dx = \frac{1}{2}\,du$

$\displaystyle = \frac{1}{2}\arcsin(2x) + C$

$\displaystyle\int \frac{1}{\sqrt{1-u^2}}\,du$

$= \arcsin u + C$

$5x^2 - 4 = 4\left(\frac{5}{4}x^2 - 1\right) = 4\left(\left(\frac{\sqrt{5}}{2}x\right)^2 - 1\right)$

Q 52 $\quad \displaystyle\int \frac{dx}{x\sqrt{5x^2-4}} = \int \frac{2}{\sqrt{5}}\,\frac{1}{\underbrace{\frac{2}{\sqrt{5}}}_{u}\sqrt{4(u^2-1)}}\,du$

$u = \frac{\sqrt{5}}{2}x \Rightarrow x = \frac{2}{\sqrt{5}}u$

$du = \frac{\sqrt{5}}{2}\,dx$

$dx = \frac{2}{\sqrt{5}}\,du$

$\displaystyle = \int \frac{1}{u\sqrt{4(u^2-1)}}\,du$

$\displaystyle\int \frac{1}{u\sqrt{u^2-1}}\,du$

$\displaystyle = \frac{1}{2}\int \frac{1}{u\sqrt{u^2-1}}\,du \rightarrow$ Use formula

$= \operatorname{arcsec} u + C$

Ex. 8.1

Q26 $\displaystyle\int \frac{6\,dy}{\sqrt{y}\,(1+y)} = \int \frac{2\cdot 6}{1+u^2}\,du = 12\int \frac{1}{1+u^2}\,du$

make it into ~~a square~~ an expression involving square

$= 12 \arctan u + C$

$= 12 \arctan \sqrt{y} + C$

$u = \sqrt{y}$

$1+y = 1+(\sqrt{y})^2$

$du = \frac{1}{2\sqrt{y}}\,dy$

$\frac{1}{\sqrt{y}}\,dy = 2\,du$

Q16

$$\int \frac{d\theta}{\sqrt{2\theta-\theta^2}} = \int \frac{d\theta}{\sqrt{1-(\theta-1)^2}}$$

$u = \theta - 1$

$du = d\theta$

$$= \int \frac{1}{\sqrt{1-u^2}}\, du$$

$1-u^2$

$$= \arcsin u + C$$

$$= \arcsin(\theta-1) + C$$

$2\theta - \theta^2$

$= -(\theta^2 - 2\theta)$

$= -(\theta^2 - 2\theta + 1 - 1)$

$= -((\theta-1)^2 - 1)$

$= 1 - (\theta-1)^2$

Q 14 $\displaystyle\int \underbrace{\csc t}_{\frac{1}{\sin t}} \sin 3t \, dt$

$\sin(3t) = \sin(2t + t)$

$= \cos 2t \sin t + \sin 2t \cos t$

$= \cos 2t \sin t + (2\sin t \cos t)\cos t$

$= \cos 2t \sin t + 2\sin t \cos^2 t$

$= (\cos 2t + 2\cos^2 t)\sin t$

$\csc t \sin(3t) = \cos 2t + \boxed{2\cos^2 t}$

$= \cos 2t + (\cos 2t + 1)$

$= 2\cos 2t + 1$

$\therefore \int \csc t \sin 3t \, dt = \sin 2t + t + C$

$\sin(nt)$

$\downarrow$

reduce it by using

$\sin(A+B) = \ldots$

$\cos^2 t$

$\downarrow$

Reduce it using

Half-Angle Formula.

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$= \cos^2\theta - (1 - \cos^2\theta)$

$= 2\cos^2\theta - 1$

Q24 $\int (\sec t + \cot t)^2 \, dt$

$s^2 + c^2 = 1$

$1 + \cot^2 t = \csc^2 t$

$\csc^2 t - 1$

$$= \int \underbrace{\sec^2 t}_{\tan t} + \underbrace{2 \sec t \cot t}_{\frac{1}{\cos t} \cdot \frac{\cos t}{\sin t} \to \csc t} + \cot^2 t \, dt$$

$$\int \sec^2 t \, dt = \tan t + C$$

$$\int \csc t \, dt = -\ln|\csc t + \cot t| + C$$

$$\int \csc^2 t \, dt = -\cot t + C$$

$$\therefore \int (\sec t + \cos t)^2 \, dt$$

$$= \tan t - 2\ln|\csc t + \cot t| - \cot t - t + C$$

Q32 $\displaystyle\int_{-1}^{1} \underbrace{\sqrt{1+x^2}\sin x}_{\text{odd}}\, dx = 0.$

Let g be an even func.

Then $\displaystyle\int_{-a}^{a} g(x)\,dx$

$\displaystyle = \boxed{2}\int_{\boxed{0}}^{a} g(x)\,dx$

make computation quickly

$\downarrow$

Fourier Series/ Fourier Transform

$f(x) := \sqrt{1+x^2}\sin x$

Let $x \in \mathbb{R}$.

$$f(-x) = \sqrt{1+(-x)^2}\sin(-x)$$

$$= \sqrt{1+x^2}\,(-\sin x)$$

$$= -\underline{\sqrt{1+x^2}\sin x}$$

$$= -f(x).$$

$\therefore f$ is an odd function.

Q6

$$\int \frac{dx}{x-\sqrt{x}} = \int \frac{dx}{\sqrt{x}(\sqrt{x}-1)} = \int \frac{2\,du}{u-1}$$

$$= 2\ln|u-1| + C$$

$$= 2\ln|\sqrt{x}-1| + C$$

$$\cancel{u = x-\sqrt{x}}$$

$$\cancel{du = 1 - \frac{1}{2\sqrt{x}}\,dx}$$

$$x-\sqrt{x} = \sqrt{x}(\sqrt{x}-1)$$

$$\frac{1}{x-\sqrt{x}} = \frac{1}{\sqrt{x}(\sqrt{x}-1)}$$

$$u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}}\,dx$$

$$\frac{1}{\sqrt{x}}\,dx = 2\,du$$

R38 $\displaystyle\int \frac{d\theta}{\cos\theta - 1}$

$\downarrow$

$\cos^2\theta$ or $\sin^2\theta$

$A^2 - B^2 = (A-B)(A+B)$

① Mult. by conjugate $\dfrac{1}{\cos\theta - 1} \cdot \dfrac{\cos\theta + 1}{\cos\theta + 1} = \dfrac{\cos\theta + 1}{\underbrace{\cos^2\theta - 1}_{-\sin^2\theta}}$

② Half angle Formula

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta \longrightarrow = (1 - \sin^2\theta) - \sin^2\theta$$

$$= \cos^2\theta - (1 - \cos^2\theta)$$

$$= 2\cos^2\theta - 1 \qquad = 1 - 2\sin^2\theta$$

$$\boxed{\cos\theta - 1} = -2\sin^2\left(\frac{\theta}{2}\right)$$

$$\boxed{\cos\theta \pm 1}$$